

BALLYKELLY, United KingdomWMO: **039080**Lat: **55.057N**Lon: **7.007W**Elev: **16**StdP: **14.69**Time Zone: **0.00 (EUW)**Period: **97-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	28.8	22.3	16.8	28.1	25.6	19.6	32.1	38.4	45.9	33.8	47.7	2.9	120	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	71.4	63.1	68.9	61.4	66.7	60.3	64.7	69.6	62.9	67.2	61.4	65.2	10.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	84.9	67.2	60.9	80.0	65.5	59.5	75.9	63.9	29.8	69.7	28.5	67.3	27.4	65.4	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.2	25.4	22.5	DB	21.7	77.3	4.9	3.2	18.2	79.6	15.3	81.4	12.6	83.2	9.0	85.5	
			WB	20.9	68.0	4.8	1.8	17.4	69.3	14.6	70.4	11.8	71.4	8.3	72.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	41.5	42.3	44.4	47.7	52.2	56.1	59.0	58.7	56.5	51.0	45.8	41.3
	DBStd	8.00	5.48	5.05	4.67	4.86	4.23	4.26	3.67	3.37	4.03	4.75	4.80	6.51
	HDD50	1249	266	218	183	101	21	3	0	0	2	46	137	272
	HDD65	5579	729	634	640	520	397	268	192	199	257	433	575	735
	CDD50	1154	2	4	8	31	89	186	278	268	196	78	12	2
	CDD65	10	0	0	0	0	0	1	6	2	1	0	0	0
	CDH74	31	0	0	0	0	0	3	24	4	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	10.0	11.5	11.6	11.0	9.6	9.6	8.8	8.5	8.3	9.4	10.6	11.3	10.4
Precipitation	PrecAvg	45.2	4.8	3.8	3.3	2.9	2.7	3.1	3.3	3.7	3.3	4.6	4.6	5.2
	PrecMax	55.5	7.0	8.5	6.7	4.5	5.3	6.5	5.6	7.5	7.2	9.2	8.7	10.6
	PrecMin	34.8	1.1	1.3	0.9	0.9	0.5	0.9	1.3	0.6	0.6	1.7	1.7	2.2
	PrecStd	5.6	1.6	1.8	1.5	1.0	1.3	1.3	1.0	1.5	1.4	1.9	1.5	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	56.6	58.8	66.2	71.3	73.7	77.2	72.6	71.9	63.5	58.7	55.8
		MCWB	52.0	54.0	52.3	54.7	60.2	63.0	66.7	64.8	64.0	58.2	55.1	52.9
	2%	DB	53.3	54.3	56.1	61.6	67.1	69.7	71.7	69.2	68.0	61.2	56.3	53.5
		MCWB	50.7	51.7	51.4	53.6	58.1	61.2	63.7	62.8	61.5	57.3	53.6	51.1
	5%	DB	51.5	52.0	53.9	58.4	63.1	66.8	68.3	66.8	65.3	59.7	54.5	51.8
		MCWB	49.0	49.5	50.0	52.3	55.9	59.9	61.5	61.3	60.0	56.1	52.1	49.4
	10%	DB	49.4	50.1	51.8	56.0	60.0	63.8	65.6	64.7	63.0	58.1	52.9	49.8
		MCWB	47.1	47.5	48.6	50.7	54.0	58.0	59.8	60.0	58.5	54.7	50.6	47.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	54.8	54.0	57.3	62.6	65.4	67.2	66.6	65.6	60.1	55.8	53.4
		MCDB	54.8	56.1	56.6	62.3	69.9	71.3	75.2	70.8	69.6	62.0	58.0	55.5
	2%	WB	51.0	51.9	52.4	54.8	59.1	62.4	64.8	64.1	62.7	58.2	54.0	51.4
		MCDB	53.1	53.7	55.1	59.6	65.4	67.7	70.3	67.5	66.2	60.5	55.9	53.2
	5%	WB	49.3	49.9	50.7	52.9	56.7	60.3	62.7	62.5	60.8	56.7	52.5	49.7
		MCDB	51.3	51.9	53.2	57.7	61.9	65.4	67.3	65.8	64.4	59.2	54.2	51.6
	10%	WB	47.3	47.8	49.0	51.1	54.8	58.8	60.8	60.8	59.1	55.2	50.8	47.7
		MCDB	49.3	49.9	51.4	55.4	59.0	63.3	64.4	63.6	62.4	57.7	52.6	49.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.6	11.4	12.5	13.0	12.1	11.7	11.7	11.4	10.2	8.5	8.9
		MCDBR	9.4	9.3	13.2	15.9	17.5	16.8	16.4	14.6	13.8	10.9	9.2	9.6
		MCWBR	8.7	8.6	9.5	10.0	10.7	10.1	9.5	9.6	9.1	8.6	8.6	9.4
	5% WB	MCDBR	9.4	9.1	11.4	14.7	15.7	15.3	14.9	12.4	12.2	10.4	8.9	9.7
		MCWBR	8.9	9.1	9.0	10.0	10.4	10.3	9.6	9.2	9.0	8.7	8.7	9.6
Clear-Sky Solar Irradiance	taub	0.344	0.353	0.374	0.385	0.391	0.380	0.383	0.367	0.363	0.349	0.334	0.332	
	taud	2.198	2.277	2.270	2.282	2.310	2.377	2.375	2.433	2.436	2.392	2.265	2.170	
	Ebn at Noon	175	226	250	265	270	275	271	269	254	226	180	149	
	Edh at Noon	20	26	32	37	38	36	35	31	27	23	18	16	
All-Sky Solar Radiation	RadAvg	170	386	723	1193	1526	1554	1434	1164	813	456	210	123	
	RadStd	18	34	63	97	135	152	108	95	44	41	21	12	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page